

Back of the Loop Estimation

Estimate + Tough exp + Common performance numbers

Power	Approximate value	Full name	Short name
10	1 Thousand	1 Kilobyte	1 KB
20	1 Million	1 Megabyte	1 MB
30	1 Billion	1 Gigabyte	1 GB
40	1 Trillion	1 Terabyte	1 TB
50	1 Quadrillion	1 Petabyte	1 PB

Latency numbers

Meet the meal prep community	Time
L1 cache reference	0.5 ns
_____ बाट में	5 ns
L2 cache reference	7 ns
_____ बाट में	100 ns
Main memory reference	100 ns
Compress 1K bytes with Zip	10,000 ns = 10 μ s
Send 2K bytes over 1 Gbps network	20,000 ns = 20 μ s
Read 1 MB sequentially from memory	250,000 ns = 250 μ s
Round trip within the same datacenter	500,000 ns = 500 μ s
Round trip to an edge server	10,000,000 ns = 10 ms

Round trip within the same datacenter	500,000 ns = 500 μ s
Disk seek	10,000,000 ns = 10 ms
Read 1 MB sequentially from the network	10,000,000 ns = 10 ms
Read 1 MB sequentially from disk	30,000,000 ns = 30 ms
Send packet CA (California) -> Netherlands -> CA	150,000,000 ns = 150 ms

After Seeing Latency ms

- Memory is faster than disk
- Avoid disk seeks if possible
- Simple compression algorithms are fast
- Compress data before sending if possible
- Data centers are far located - Time taking

III Availability no. -

- High avail = systems continuously operational for long
- Measured in %

100% = 0 downtime

III SLA (Service Level Agreement)

- Agreement between service provider.

III Twitter - IOPS and Storage Requirement.

T ...

... active users - 200 million

Interview Q

Few Assumptions

Estimation

Monthly active users = 300 million

Daily users = 50%

Tweets / day on Avg = 2

Tweets containing media = 10%

Tweets stored $\rightarrow$ 5 yrs

Daily Active users $\rightarrow$ 50% of 300 M

= 150 M

Tweets QPS $\rightarrow$ 150 M * 2 tweets / 24 hr / 3600 sec

$$= \sqrt{\sim 3500}$$

Peak QPS = 2 * QPS = $\sim$ 7000

(10% users)

Storage Req Est: -

Avg Tweet Size

* tweet id = 64 bytes

* text = 150 bytes

* media = 1 MB

media storage = (Per day active users) * (No. of Tweets)

* (10% contain media)

* Size of media.

$$= 150 \text{ M} * (2) * (10\%) * 1 \text{ MB}$$

$$= 150 \text{ M} * (2) * (10\%) * 1 \text{ MB}$$

$$= 30 \times 10^6 \text{ MB}$$

$$= 30 \text{ TB/day}$$

$$\begin{aligned} \textcircled{*} 5 \text{ yrs} &\rightarrow 30 \text{ TB} \\ &\times 5 + 365 \\ &= 54,750 \text{ TB} \\ &\approx \underline{\underline{55 \text{ TB}}} \end{aligned}$$